

RADU ANDREI TUDORICA

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EDUCATION

National University of Science and Technology POLITEHNICA Bucharest Sept 2024 – Present
Bachelor of Science in Computer Science (ACS CTI) Bucharest, Romania

- **Relevant Coursework:** Algorithm Analysis, Data Structures, Java, Scala, Functional Programming, Computer Networks, Communication Protocols, Operating Systems, Assembly Language & Computer Architecture, Numerical Methods

EXPERIENCE

Google Summer of Code 2026 – Open-Source Contributor May 2026 – Sept 2026
Unikraft Remote

- Ported and updated 6 application library ports (Redis, Nginx, SQLite, Lua, Python3, libgo) to the current Unikraft unikernel core, resolving build failures, patch conflicts, and missing syscalls.
- Diagnosed and fixed concrete build blockers: rewrote patches for changed function signatures, added missing source files and a C++ dependency to the build system, and confirmed successful QEMU boot.
- Contributed 3 merged PRs to Unikraft prior to GSoC, covering unit tests and error handling for memory allocator subsystems.

PROJECTS

BugTrackerPro – Issue Management Backend | *Java, OOP, Design Patterns, JSON* | [GitHub](#) Dec 2025 – Jan 2026

- Designed the backend architecture for a ticket management system simulating real-world agile workflows with milestones, assignments, and status tracking.
- Implemented the **Command Pattern** to handle 20+ distinct system actions, achieving high cohesion and loose coupling between modules.
- Built a robust filtering engine using the **Strategy Pattern** to query an in-memory database with complex, composable search criteria.
- Developed automated reporting modules computing resolution efficiency, customer impact scores, and system stability metrics.

Lox Interpreter – Java Implementation | *Java, Compiler Architecture, AST, Visitor Pattern* | [GitHub](#) Jul 2025 – Present

- Built a complete interpreter from scratch: lexical scanner, recursive descent parser, and tree-walk evaluator.
- Implemented runtime support for variable scoping, first-class functions, closures, and control flow statements.
- Automated AST node class generation via a custom Java metaprogramming tool, reducing boilerplate code.
- Applied the **Visitor Pattern** for clean separation between syntax tree structure and evaluation logic.

EAN-13 Barcode Decoder & In-Memory Database | *Scala, Functional Programming* | [GitHub](#) Apr 2026

- Implemented a purely functional EAN-13 barcode decoder processing raw PPM bitmap images – zero mutable state (val-only), using run-length encoding and rational arithmetic.
- Built a type-safe in-memory relational database with CSV parsing, filtering, sorting, and CRUD operations.
- Designed a composable SQL-like query DSL using implicit conversions, algebraic data types, and higher-order functions.

TECHNICAL SKILLS

Languages: Java, Scala, Python, C/C++, Bash
Frameworks & Libraries: Maven, Jackson (JSON), JUnit
Developer Tools: Git, GitHub, IntelliJ IDEA, VS Code, Linux, Remote Access Tools
Concepts: OOP, Design Patterns, Functional Programming, Compiler Design, Algorithms & Data Structures